INCH-POUND

The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 November 2022.

MIL-PRF-19500/411T
w/AMENDMENT 2
21 July 2022
SUPERSEDING
MIL-PRF-19500/411T
w/AMENDMENT 1
24 September 2021

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY, ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT PACKAGES), AND UN-ENCAPSULATED, DEVICE TYPES 1N5415 THROUGH 1N5420, QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for fast recovery silicon rectifier diodes. Four levels of product assurance (JAN, JANTX, JANTXV and JANS) are provided for each encapsulated device. Two levels of product assurance (JANHC and JANKC) are provided for each un encapsulated device type. Provision for radiation hardness assurance (RHA) to four radiation ("M", "D", "R", and "H") test levels is provided for JANTXV and JANS product assurance levels.

1.2 Package outlines. The device package outline is as follows: axial lead in accordance with figure 1 and surface mount in accordance with figure 2 for all encapsulated device types. The dimensions, physical characteristics and topography for JANHC and JANKC un-encapsulated die are in accordance with figures 3.

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = 25°C.

| Col. 1 | Col. 2 | Col. 3 | Col. 4 | Col. 5 | Col. 6 | Col. 7 | Col. 8 | Col. 9 | Col. 10 | Col. 11 |
|---|---|---|---|---|---|---|---|---|---|---|
| Types | $V_{BR}$ | $V_{RWM}$ | $I_O$ (1) $T_A$ = 55°C (2) (3) | $I_O$ $T_A$ = 100°C (2) (3) | $I_{FSM}$ $I_O$ = 2 A dc $T_A$ = 55°C tp = 8.3 ms | $t_{rr}$ | $T_{STG}$ and $T_J$ | $R_{\theta JL}$ (4) at L = .375 in. (9.53 mm) | $R_{\theta JEC}$ (5) at L = 0 for US versions | $R_{\theta JX}$ (3) |
| | V dc | V (pk) | A dc | A dc | A (pk) | ns | °C | °C/W | °C/W | °C/W |
| 1N5415,US | 50 | 50 | 3 | 2 | 80 | 150 | -65 | 22 | 6.5 | 43 |
| 1N5416,US | 100 | 100 | 3 | 2 | 80 | 150 | to | 22 | 6.5 | 43 |
| 1N5417,US | 200 | 200 | 3 | 2 | 80 | 150 | +175 | 22 | 6.5 | 43 |
| 1N5418,US | 400 | 400 | 3 | 2 | 80 | 150 | | 22 | 6.5 | 43 |
| 1N5419,US | 500 | 500 | 3 | 2 | 80 | 250 | | 22 | 6.5 | 43 |
| 1N5420,US | 600 | 600 | 3 | 2 | 80 | 400 | | 22 | 6.5 | 43 |

See notes on next page.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Maximum ratings – continued.

(1) Derate linearly at 22 mA/°C for 55°C ≤ $T_A$ ≤ 100°C.
(2) Derate linearly at 26.7 mA/°C for 100°C ≤ $T_A$ ≤ 175°C.
(3) For the 3A rating at 55°C ambient and the 2 A rating at 100°C ambient, these $I_O$ ratings are for a thermally mounting methods (PC boards or other) where the lead or end-cap temperatures cannot be maintained and where thermal resistance from mounting point to ambient is still sufficiently controlled where $T_{J(MAX)}$ in 1.3 is not exceeded. This equates to $R_{\theta JX}$ ≤ 43°C/W in col. 11 of 1.3. Also see application notes in 6.6.
(4) See figure 4.
(5) See figure 5.

1.4 Primary electrical characteristics. Not applicable.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "R", and "H".

1.5.3 Device type. The designation system for the device types of diodes covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The diodes of this specification sheet use the first number and letter symbols "1N".

1.5.3.2 Second number symbols. The second number symbols for the diodes covered by this specification sheet are as follows: "5415", "5416", "5417", "5418", "5419" and "5420".

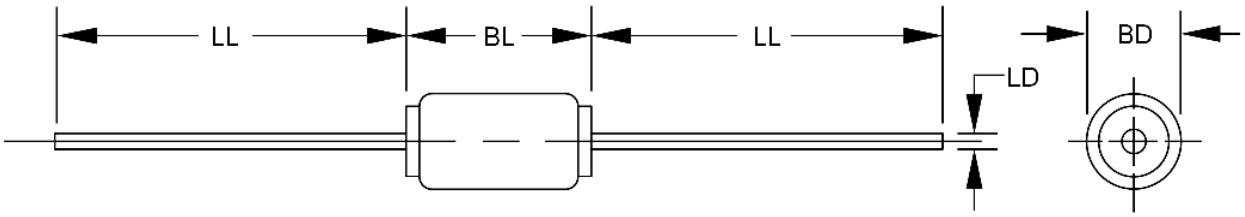
1.5.4 Suffix letters. No suffix letters are used on devices that are packaged in the axial leaded package of figure 1. The suffix letters "US" are used on devices that are packaged in the surface mount package of figure 2.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet is "A" (see figures 3 and 6.5.2).



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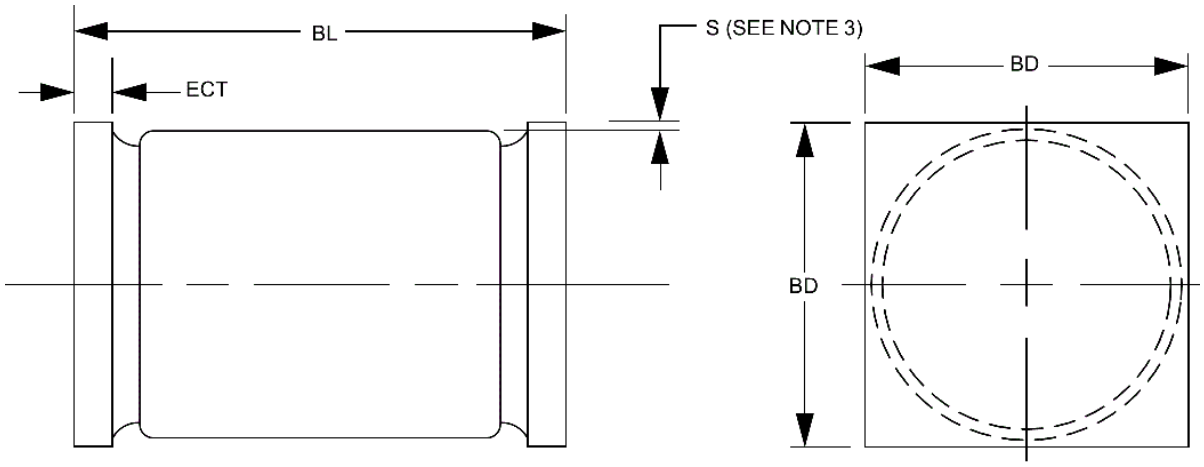

| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | Inches | mm | mm | |
| | Min | Max | Min | Max | |
| BD | .110 | .180 | 2.79 | 4.57 | 3 |
| LD | .036 | .042 | 0.91 | 1.07 | 4 |
| BL | .130 | .260 | 3.30 | 6.60 | 4 |
| LL | .90 | 1.30 | 22.9 | 33.0 | |

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension BD shall be measured at the largest diameter.
4. The BL dimension shall include the entire body including slugs and sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.
5. In accordance with ASME Y14.5, diameters are equivalent to øx symbology.

FIGURE 1. Physical dimensions of axial leaded package.



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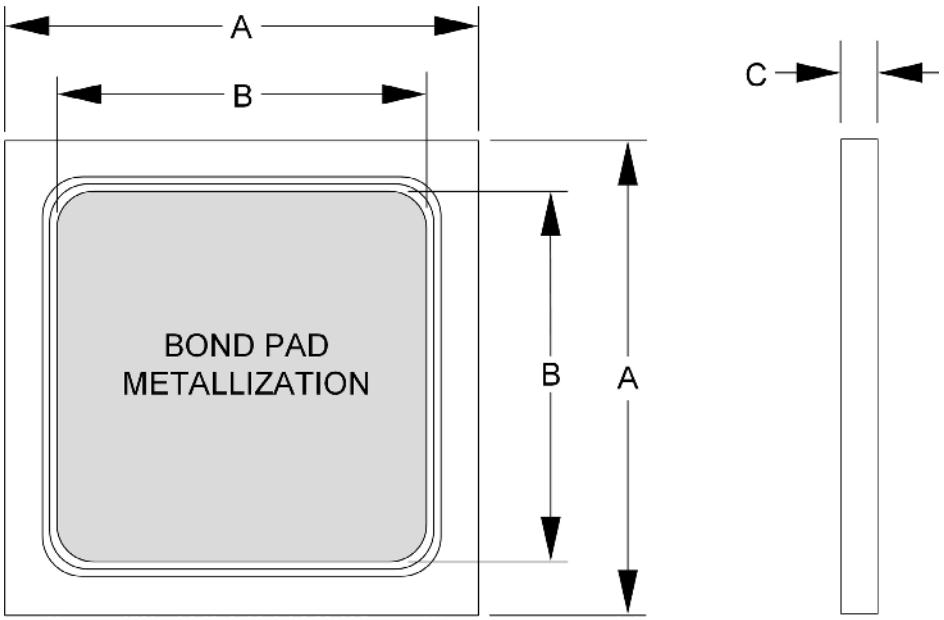

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | Inches | mm | mm |
| | Min | Max | Min | Max |
| BL | .200 | .225 | 5.080 | 5.72 |
| BD | .137 | .148 | 3.48 | 3.76 |
| ECT | .019 | .028 | 0.48 | 0.71 |
| S | .003 | | 0.08 | |

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Dimensions are pre-solder dip.
3. Minimum clearance of glass body to mounting surface on all orientations.
4. In accordance with ASME Y14.5, diameters are equivalent to øx symbology.

FIGURE 2. Physical dimensions of square end-cap surface mount (US) package.



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| | Dimensions | | | |
|---|---|---|---|---|
| Letter | Inches | Inches | mm | mm |
| | Min | Max | Min | Max |
| A | 0.085 | 0.095 | 2.16 | 2.41 |
| B | 0.063 | 0.073 | 1.60 | 1.85 |
| C | 0.006 | 0.016 | 0.15 | 0.41 |

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.

FIGURE 3. JANKCA and JANHCA (A-version) die dimensions.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$P_{RM}$ Peak reverse power

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (axial lead), figure 2 (surface mount), and figure 3 for JANHC and JANKC die.

3.4.1 Lead finish. Unless otherwise specified, lead or end-cap finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for finish the maximum lead temperature is limited to +175°C maximum. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature metallurgical bonding between both sides of the silicon die and terminal pins. Metallurgical bond shall be in accordance with the requirements of category I in MIL-PRF-19500.



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3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking of US version. For US version only, all marking may be omitted from the device except for the cathode marking. All marking which is omitted from the body of the device shall appear on the label of the initial container.

3.5.2 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end. Alternately for US surface mount devices, a minimum of three evenly spaced contrasting color dots around the periphery of the cathode end may be used. No color coding will be permitted.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics shall be as specified in 1.3 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7.1 Post-irradiation performance characteristics. The electrical performance characteristics of the RHA devices shall be as specified in 4.4.4 herein.

3.8 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, part number designators, and test levels shall be as defined in MIL-PRF-19500.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation of un-encapsulated die (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Qualification for radiation hardness assurance. Qualification inspection for radiation hardness assured JANS and JANTXV devices shall consist of group D examinations and tests specified in table II herein.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.3 Qualification of un-encapsulated die (JANHC and JANKC). Qualification inspection for unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500 and as specified herein.



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4.3 Screening (quality levels JANS, JANTXV and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screening | Measurements for JANS level | Measurements for JANTX and JANTXV levels |
|---|---|---|
| 2 | Optional | Not required |
| (1) 3c | Thermal impedance (see 4.3.1) | Thermal impedance (see 4.3.1) |
| 9 | $I_{R1}$ and $V_{F1}$ | Not required |
| 10 | Method 1038 of MIL-STD-750, condition A | Method 1038 of MIL-STD-750, condition A |
| (2) 11 | $I_{R1}$ and $V_{F1}$;<br>$\Delta I_{R1}$ ≤ 100 percent of initial reading or 250 nA dc, whichever is greater;<br>$\Delta V_{F1}$ ≤ ±0.1 V dc | $I_{R1}$ and $V_{F1}$ |
| 12 | Required, see 4.3.2 | Required, see 4.3.2 |
| (2) (3) 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{R1}$ ≤ 100 percent of initial value or 250 nA dc, whichever is greater;<br>$\Delta V_{F1}$ ≤ ±0.1 V dc,<br>scope display evaluation (see 4.5.3) | Subgroup 2 of table I herein;<br>$\Delta I_{R1}$ ≤ 100 percent of initial value or 250 nA dc, whichever is greater;<br>$\Delta V_{F1}$ ≤ ±0.1 V dc,<br>scope display evaluation (see 4.5.3) |
| 15 | Required | Not required |
| 16 | Required | Not required |

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) For JANTX and JANTXV devices, $\Delta V_{F1}$ may be omitted if thermal impedance is performed, unless irradiation is used to reduce the carrier lifetime.
(3) $Z_{\theta JX}$ is not required in screen 13, if already previously performed.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$ and $t_{MD}$ shall be 10 ms. The thermal impedance limit shall comply with the thermal impedance graphs herein (less than or equal to the curve value at the same $t_H$ time) and shall be less than the process determined statistical maximum limit as outlined in method 3101. See table III, subgroup 4 herein.

4.3.2 Power burn-in conditions. Power burn-in shall be performed in accordance with test condition B of method 1038 of MIL-STD-750. The details of 4.5.2 and those that follows shall apply: $I_O$ = 3 A minimum; $T_A$ = 55°C maximum. Adjust $I_O$ or $T_A$ to achieve the required $T_J$. $T_J$ = 135°C minimum. With approval of the qualifying activity, alternate burn-in criteria (hours, bias conditions, $T_J$, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing sites burn-in data and performance history will be essential criteria for burn-in modification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. The $Z_{\theta JX}$ end-point shall be derived by the supplier and approved by the qualifying activity. This $Z_{\theta JX}$ end-point shall be documented in the qualification report.



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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Delta measurements shall be as specified in table IV herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500. For B5, if a failure occurs, resubmission shall be at the test conditions of the original sample.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 4066 | Condition A, $I_{FSM}$ = rated $I_{FSM}$ (see col. 6 of 1.3); ten surges of 8.3 ms each at 1 minute intervals, superimposed on $I_O$ = 2 A dc, $V_{RWM}$ = rated $V_{RWM}$ (see col. 3 of 1.3). |
| B4 | 1037 | See 4.5.2. Adjust $I_O$ to achieve the required $T_J$, apply (see col. 5 of 1.3), $V_R$ = rated $V_{RWM}$ (see col. 3 of 1.3); 2,000 cycles. |
| B5 | 1027 | See 4.5.2. $I_O$ = 3 A minimum (see col. 4 of 1.3); apply $V_R$ = rated $V_{RWM}$ (see col. 3 of 1.3), adjust $I_O$ or $T_A$ to achieve $T_J$ = 175 °C minimum; f = 50 to 60 Hz; $T_A$ = 55 °C max. t = 1,000 hours; n = 45, c = 0. For irradiated devices, include $t_{rr}$ as an end-point measurement. |
| B8 | 4065 | $P_{RM}$ ≥ 636 W for square wave in accordance with test method 4065 of MIL-STD-750 ($P_{RM}$ ≥ 1,000 W for half-sine wave). Test shall be performed on each sublot; sampling plan n = 10, c = 0, electrical end-points, see table I, subgroup 2 herein. |

4.4.2.2 Quality levels JAN, JANTX, and JANTXV, table E-VIB of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1027 | See 4.5.2. $I_O$ = 3 A minimum (see col. 4 of 1.3); adjust $I_O$ or $T_A$ to achieve the required $T_J$. Apply $V_R$ = rated $V_{RWM}$ (see col. 3 of 1.3), f = 50 to 60 Hz. $T_A$ = 55 °C maximum. $T_J$ = 150 °C minimum. For irradiated devices, include $t_{rr}$ as an end-point measurement. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Delta measurements shall be as specified in table IV herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Axial devices only: Tension: Test condition A; weight = 20 pounds (9.1 kg), t = 15 s. Lead fatigue: Test condition E; weight = 2 pounds (0.91 kg). |
| | 2038 | US devices only: Tab pull: Test condition B; weight = 20 pounds (9.1 kg), t = 15 seconds. |
| C5 | 4081 | See 4.5.4. For axial devices; $R_{\theta JL}$ (maximum) ≤ 22 °C/W; L = .375 inch (9.53 mm). For US version surface mount devices; $R_{\theta JEC}$ = 6.5 °C/W maximum. |
| C6 | 1026 | See 4.5.2. $I_O$ = 3 A minimum (see col. 4 of 1.3); adjust $I_O$ or $T_A$ to achieve the required $T_J$; apply $V_R$ = rated $V_{RWM}$ (see col. 3 of 1.3), f = 50 to 60 Hz, $T_A$ = 55 °C maximum, $T_J$ = 150 °C minimum. For irradiated devices, include $t_{rr}$ as an end-point measurement. |



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4.4.4 Group D inspection. Radiation hardness assured JANS and JANTXV devices shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose or method 1017 for neutron fluence as applicable.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein. See table IV for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

4.5.2.1 Burn-in. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the required $T_J$ and $I_O$ and that the minimum required voltage, where applicable, is maintained through-out the burn-in period.

4.5.2.2 Junction temperature measurements. Use method 3100 of MIL-STD-750 to measure $T_J$.

4.5.3 Scope display evaluation. Scope display evaluation shall be stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only, with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. The reverse current ($I_{BR}$) over the knee shall be 500 µA peak.

4.5.4 Thermal resistance. Thermal resistance measurement shall be performed in accordance with method 4081 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, and $t_H$. Measurement delay time $t_{MD}$ = 70 µs max. See MIL-PRF-19500, table E-IX, subgroup 4. Forced moving air or draft shall not be permitted across the devices during test.

4.5.5 Peak reverse power characterization. The testing to characterize $P_{RM}$ shall be conducted in accordance with method 4065 of MIL-STD-750. The following details shall apply:

a. During the test, the voltage ($V_{BR}$) shall be monitored to verify it has not collapsed.

b. Any collapse in $V_{BR}$ during or after the test, or rise in leakage current ($I_R$) after the test that exceeds $I_{R1}$ in table I, shall be considered a failure to that level of applied $P_{RM}$.

c. Progressively higher levels of $P_{RM}$ shall be applied until failure occurs on all devices within the chosen sample size to characterize each sublot.

4.6 Element evaluation of un-encapsulated die. The element evaluation of un-encapsulated die shall be in accordance with appendix G of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3101 | See 4.3.1 | $Z_{\theta JX}$ | | | °C/W |
| Forward voltage | 4011 | Condition A, $I_F$ = 1.5 A dc (pulsed see 4.5.1) | $V_{F1}$ | 0.5 | 1.2 | V |
| Forward voltage | 4011 | Condition A, $I_F$ = 9 A dc (pulsed see 4.5.1); tp = 300 µs; 2 percent maximum duty cycle | $V_{F2}$ | 0.6 | 1.5 | |
| Reverse current | 4016 | Condition A or B, $V_R$ = rated (see col. 2 of 1.3). | $I_{R1}$ | | 1.0 | µA |
| Breakdown voltage | 4021 | $I_R$ = 50 µA dc | $V_{(BR)1}$ | | | |
| 1N5415, 1N5415US | | | | 55 | | V |
| 1N5416, 1N5416US | | | | 110 | | V |
| 1N5417, 1N5417US | | | | 220 | | V |
| 1N5418, 1N5418US | | | | 440 | | V |
| 1N5419, 1N5419US | | | | 550 | | V |
| 1N5420, 1N5420US | | | | 660 | | V |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = + 100°C | | | | |
| Reverse current | 4016 | Condition A or B, $V_R$ = rated (see col. 2 of 1.3). | $I_{R2}$ | | 20 | µA |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Forward voltage | 4011 | Condition A, $I_F$ = 0.5 A (pulsed see 4.5.1) | $V_{F3}$ | 0.5 | 1.4 | V |
| Breakdown voltage | 4021 | $I_R$ = 50 µA | $V_{(BR)2}$ | | | |
| 1N5415, 1N5415US | | | | 50 | | V |
| 1N5416, 1N5416US | | | | 100 | | V |
| 1N5417, 1N5417US | | | | 200 | | V |
| 1N5418, 1N5418US | | | | 400 | | V |
| 1N5419, 1N5419US | | | | 500 | | V |
| 1N5420, 1N5420US | | | | 600 | | V |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 4 | | | | | | |
| Reverse recovery time | 4031 | Condition B1 | $t_{rr}$ | | | |
| 1N5415, 1N5415US | | | | | 150 | ns |
| 1N5416, 1N5416US | | | | | 150 | ns |
| 1N5417, 1N5417US | | | | | 150 | ns |
| 1N5418, 1N5418US | | | | | 150 | ns |
| 1N5419, 1N5419US | | | | | 250 | ns |
| 1N5420, 1N5420US | | | | | 400 | ns |
| Capacitance | 4001 | $V_R$ = 4 V dc; 100 kHz ≤ f ≤ 1 MHz | C | | | |
| 1N5415, 1N5415US | | | | | 550 | pF |
| 1N5416, 1N5416US | | | | | 430 | pF |
| 1N5417, 1N5417US | | | | | 250 | pF |
| 1N5418, 1N5418US | | | | | 165 | pF |
| 1N5419, 1N5419US | | | | | 140 | pF |
| 1N5420, 1N5420US | | | | | 120 | pF |
| Scope display evaluation | 4023 | See 4.5.3, n = 116, c = 0 | | | | |
| Subgroup 5 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 6 | | | | | | |
| Forward surge current | 4066 | Condition A, $I_{FSM}$ = rated (see col. 6 of 1.3); ten surges of 8.3 ms each at 1 minute intervals, superimposed on $I_O$ = 2 A dc, $V_{RWM}$ = rated $V_{RWM}$ (see col. 3 of 1.3). | | | | |
| Electrical measurement | | See table IV, step 1. | | | | |
| Subgroup 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



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TABLE II. Group D inspection and end-point limits for RHA JANS and JANTXV devices only.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Neutron irradiation | 1017 | | | | | |
| Electrical measurements | | $I_F$ = 9 A dc, (pulsed see 4.5.1) | | | | |
| Forward voltage, M, D, R, H | 4011 | Condition A, tp = 300 µs; 2 percent maximum duty cycle | $V_F$ | | | |
| 1N5415, 1N5415US | | | | 0.6 | 1.6 | V |
| 1N5416, 1N5416US | | | | 0.6 | 1.6 | V |
| 1N5417, 1N5417US | | | | 0.6 | 1.7 | V |
| 1N5418, 1N5418US | | | | 0.6 | 1.7 | V |
| 1N5419, 1N5419US | | | | 0.6 | 1.8 | V |
| 1N5420, 1N5420US | | | | 0.6 | 1.8 | V |
| Reverse current, M, D, R, H | 4016 | Condition A or B | $I_R$ | | | |
| 1N5415, 1N5415US | | $V_R$ = 50 V | | | 1.0 | µA |
| 1N5416, 1N5416US | | $V_R$ = 100 V | | | 1.0 | µA |
| 1N5417, 1N5417US | | $V_R$ = 200 V | | | 1.0 | µA |
| 1N5418, 1N5418US | | $V_R$ = 400 V | | | 1.0 | µA |
| 1N5419, 1N5419US | | $V_R$ = 500 V | | | 1.0 | µA |
| 1N5420, 1N5420US | | $V_R$ = 600 V | | | 1.0 | µA |
| Subgroup 2 | | | | | | |
| Total dose irradiation | 1019 | Condition A | | | | |
| Electrical measurements | | $I_F$ = 9 A dc (pulsed see 4.5.1) | $V_F$ | | | |
| Forward voltage, M, D, R. H | 4011 | Condition A, tp = 300 µs; 2 percent maximum duty cycle | | | | |
| 1N5415, 1N5415US | | | | 0.6 | 1.6 | V |
| 1N5416, 1N5416US | | | | 0.6 | 1.6 | V |
| 1N5417, 1N5417US | | | | 0.6 | 1.7 | V |
| 1N5418, 1N5418US | | | | 0.6 | 1.7 | V |
| 1N5419, 1N5419US | | | | 0.6 | 1.8 | V |
| 1N5420, 1N5420US | | | | 0.6 | 1.8 | V |

See footnote at end of table.



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TABLE II. Group D inspection and end-point limits for RHA JANS and JANTXV devices only – Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Reverse current, M, D, R, H | 4016 | Condition A or B | $I_R$ | | | |
| 1N5415, 1N5415US | | $V_R$ = 50 V dc | | | 1.0 | µA |
| 1N5416, 1N5416US | | $V_R$ = 100 V dc | | | 1.0 | µA |
| 1N5417, 1N5417US | | $V_R$ = 200 V dc | | | 1.0 | µA |
| 1N5418, 1N5418US | | $V_R$ = 400 V dc | | | 1.0 | µA |
| 1N5419, 1N5419US | | $V_R$ = 500 V dc | | | 1.0 | µA |
| 1N5420, 1N5420US | | $V_R$ = 600 V dc | | | 1.0 | µA |

1/ For sampling plan, see MIL-PRF-19500.

TABLE III. Group E inspection (all quality levels) for qualification and requalification only.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1A | | | 45 devices, c = 0 |
| Temperature cycling (air to air) | 1051 1/ | 20 cycles, low temperature shall be -195°C and high temperature shall be +150°C. | |
| Hermetic seal 2/ | 1071 | Fine and gross as applicable. | |
| Electrical measurements | | See table I, subgroup 2, and table IV, herein. | |
| Subgroup 1B | | | |
| Temperature cycling (air to air) | 1051 | 500 cycles, condition C, -65°C to +175°C. | |
| Hermetic seal 2/ | 1071 | Fine and gross as applicable. | |
| Electrical measurements | | See table IV, herein. | |
| Subgroup 2 | | | |
| Steady-state dc blocking life | 1048 | 1,000 hours, $V_R = V_{RWM}$ (see col. 3 of 1.3). | |
| Electrical measurements | | See table I, subgroup 2, and step 1 of table IV. For irradiated devices, include $t_{rr}$ as an end-point measurement. | |

See footnotes at end of table.



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TABLE III. Group E inspection (all quality levels) for qualification and requalification only - Continued.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 4 | | | |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 5 | | | 22 devices, c = 0 |
| Barometric pressure, (reduced) | 1001 | Condition D, (pressure 8.0 mm). | |
| Subgroup 8 3/ | | | 45 devices |
| Peak reverse power | 4065 | $P_{RM}$ shall be characterized by the supplier and this data shall be available to the Government. Test shall be performed on each sublot. | |
| Electrical measurements | | See 4.5.5. | |
| Subgroup 9 | | | 45 devices |
| Resistance to glass cracking | 1057 | Step stress to destruction by increasing cycles or up to a maximum of 25 cycles. | |
| Subgroup 10 | | | 22 devices, c = 0 |
| Forward surge current | 4066 | Condition A, $I_{FSM}$ = 80 A(pk); ten surges of 8.3 ms each at 1 minute intervals, superimposed on $I_O$ = 2 A dc; $V_{RWM}$ = rated $V_{RWM}$ (see col. 3 of 1.3). $T_A$ = +55°C. | |
| Electrical measurements | | See table I, subgroup 2, and table IV, herein. | |

1/ Test method 1056 condition D, using liquid nitrogen may be used in lieu of TM 1051.
2/ Opaque glass double plug non cavity axial lead diodes may use method 2068 in lieu of method 1071.
3/ The sample size for this step stress requirement shall be determined by the supplier. A statistically significant sample size is required.



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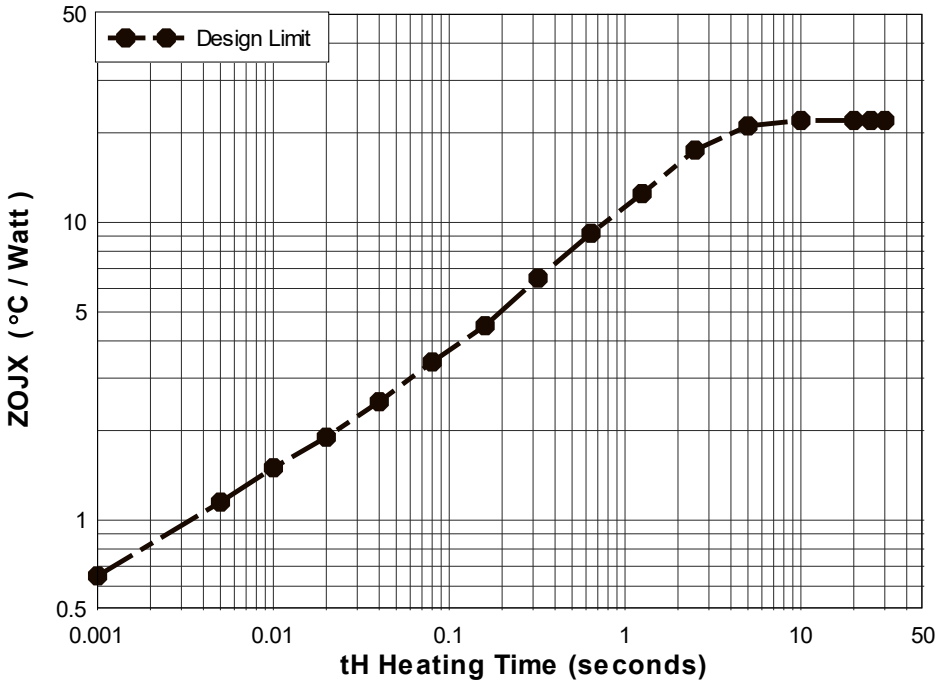
TABLE IV. Delta measurements. 1/ 2/ 3/ 4/ 5/

| Step | Inspection | MIL-STD-750 | | Symbol | Limit |
|---|---|---|---|---|---|
| | | Method | Conditions | | |
| 1. | Reverse current | 4016 | Condition A or B<br>$V_R$ = rated (see col. 2 of 1.3) | $\Delta I_R$ | 100 percent of initial value or 250 nA dc, whichever is greater. |

1/ Devices which exceed the table I, subgroup 2 (group A) limits for this test shall not be accepted.
2/ The delta measurements taken during group B inspection for quality level JANS (table E-VIA of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 4, the measurements of step 1 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 5, the measurements of step 1 of this table shall also be taken.
3/ The delta measurements taken during group B inspection for quality levels JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of step 1 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of step 1 of this table shall also be taken.
4/ The delta measurements taken during group C inspection for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of step 1 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of step 1 of this table shall also be taken.
5/ The delta measurements taken during group E inspection for all quality levels (table E-IX of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 1, the measurements of step 1 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 2, the measurements of step 1 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 10, the measurements of step 1 of this table shall also be taken.



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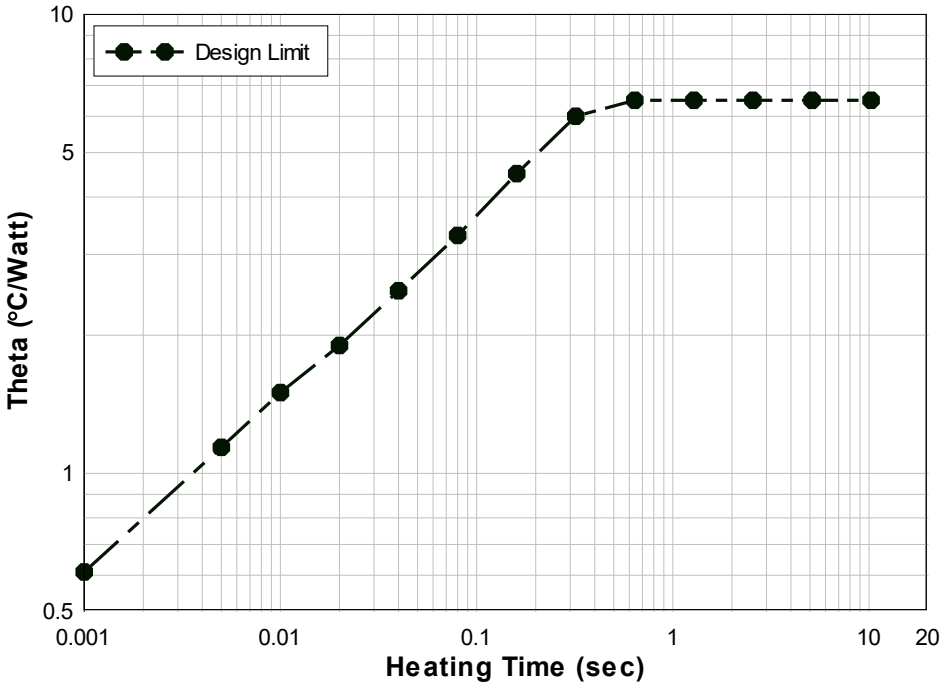

$Z_{\theta JX}$ = 1.5°C/W at 10 ms

$R_{\theta JL}$ = 22°C/W

FIGURE 4. Axial leaded thermal-impedance curve, maximum.



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$Z_{\theta JX}$ = 1.5°C/W at 10 ms

$R_{\theta JEC}$ = 6.5°C/W

FIGURE 5. Surface mount thermal-impedance curve, maximum.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

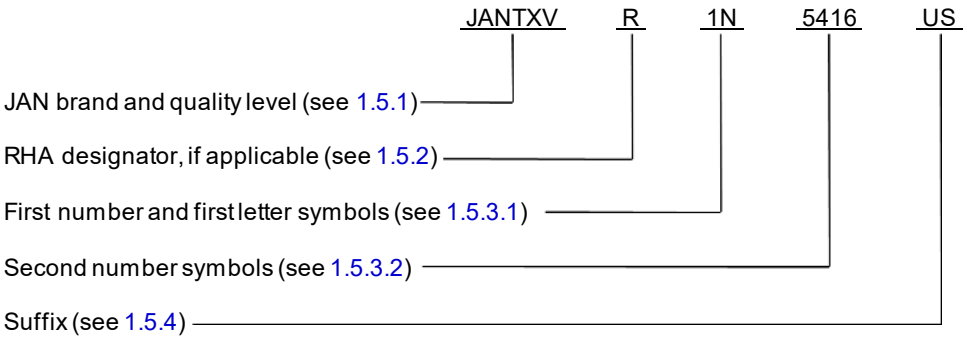
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction examples.

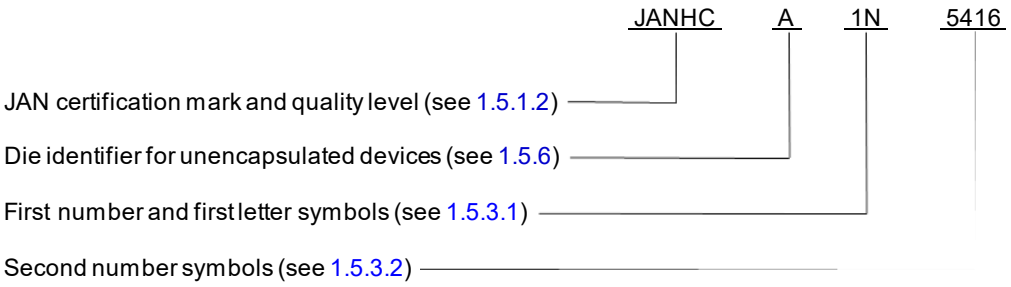
6.4.1 Encapsulated devices. The PINs for encapsulated devices are construction using the following form.






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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.




6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs available on this specification sheet.

Axial lead package of figure 1

| PINs for devices of the "JAN" quality level | PINs for devices of the "JANTX" quality level | PINs for devices of the "JANTXV" quality level | PINs for devices of the "JANTXV" quality level w/RHA (1) | PINs for devices of the "JANS" quality level | PINs for devices of the "JANS" quality level w/RHA (1) |
|---|---|---|---|---|---|
| JAN1N5415 | JANTX1N5415 | JANTXV1N5415 | JANTXV#1N5415 | JANS1N5415 | JANS#1N5415 |
| JAN1N5416 | JANTX1N5416 | JANTXV1N5416 | JANTXV#1N5416 | JANS1N5416 | JANS#1N5416 |
| JAN1N5417 | JANTX1N5417 | JANTXV1N5417 | JANTXV#1N5417 | JANS1N5417 | JANS#1N5417 |
| JAN1N5418 | JANTX1N5418 | JANTXV1N5418 | JANTXV#1N5418 | JANS1N5418 | JANS#1N5418 |
| JAN1N5419 | JANTX1N5419 | JANTXV1N5419 | JANTXV#1N5419 | JANS1N5419 | JANS#1N5419 |
| JAN1N5420 | JANTX1N5420 | JANTXV1N5420 | JANTXV#1N5420 | JANS1N5420 | JANS#1N5420 |

(1) The number sign (#) represents one of four RHA designators available on this specification sheet ("M", "D", "R", or "H").

Surface mount package of figure 2

| PINs for devices of the "JAN" quality level | PINs for devices of the "JANTX" quality level | PINs for devices of the "JANTXV" quality level | PINs for devices of the "JANTXV" quality level w/RHA (1) | PINs for devices of the "JANS" quality level | PINs for devices of the "JANS" quality level w/RHA (1) |
|---|---|---|---|---|---|
| JAN1N5415US | JANTX1N5415US | JANTXV1N5415US | JANTXV#1N5415US | JANS1N5415US | JANS#1N5415US |
| JAN1N5416US | JANTX1N5416US | JANTXV1N5416US | JANTXV#1N5416US | JANS1N5416US | JANS#1N5416US |
| JAN1N5417US | JANTX1N5417US | JANTXV1N5417US | JANTXV#1N5417US | JANS1N5417US | JANS#1N5417US |
| JAN1N5418US | JANTX1N5418US | JANTXV1N5418US | JANTXV#1N5418US | JANS1N5418US | JANS#1N5418US |
| JAN1N5419US | JANTX1N5419US | JANTXV1N5419US | JANTXV#1N5419US | JANS1N5419US | JANS#1N5419US |
| JAN1N5420US | JANTX1N5420US | JANTXV1N5420US | JANTXV#1N5420US | JANS1N5420US | JANS#1N5420US |

(1) The number sign (#) represents one of four RHA designators available on this specification sheet ("M", "D", "R", or "H").



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6.5.2 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet for unencapsulated die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCA1N5415) will be identified on the QML.

| PIN | Manufacturer |
|---|---|
| | 34156 |
| JAN1N5415 | JANHCA1N5415,<br>JANKCA1N5415 |
| JAN1N5416 | JANHCA1N5416<br>JANKCA1N5416 |
| JAN1N5417 | JANHCA1N5417<br>JANKCA1N5417 |
| JAN1N5418 | JANHCA1N5418<br>JANKCA1N5418 |
| JAN1N5419 | JANHCA1N5419<br>JANKCA1N5419 |
| JAN1N5420 | JANHCA1N5420<br>JANKCA1N5420 |



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6.6 Application data.

6.6.1 PCB mounting with FR4 material for full 3 amp $I_O$. For a PCB mounting example with FR4 material where the full 3 amp $I_O$ rating (half-sine-wave) is used at a $T_J$ of 175°C and ambient temperature of 55°C, the following steps guide the user in what the PCB pad size will need to be with 1 oz, 2 oz, and 3 oz copper. For axial-leaded, the lead length for mounting will be .187 inch (4.76 mm) or less from body to entry point on PCB surface.

a. Use the $I_O$ versus $P_O$ curve on figure 6 to look up 3 amps (X-axis) and follow up to the $T_J$ = +175°C curve (lower) for 2.81 watts.

b. Calculate maximum thermal resistance needed (+175°C - 55°C) / 2.81 W = 43°C/W.

c. Look up thermal resistance of 43°C/W on Y-axis using a thermal resistance versus pad area plot on one of the three curves on figure 7 for different weights of copper cladding and then intersect curve horizontally to get answer. These curves assume still air, horizontal position.

d. In this example, the copper mounting pad sizes for the different copper foil weights would be as follows:

(1) 1.00 $in^2$ (645.1 $mm^2$) for 1 ounce copper foil.
(2) .50 $in^2$ (322.6 $mm^2$) for 2 ounce copper foil.
(3) .32 $in^2$ (206.5 $mm^2$) for 3 ounce copper foil.

e. Add a conservative guard-band to the pad size (larger) to keep $T_J$ below +175°C.

6.6.2 PCB mounting with FR4 material for 1 amp $I_O$. For a PCB mounting example with FR4 material to support a 1 amp $I_O$ square wave switching at a 0.50 duty factor (50 percent duty cycle) at $T_J$ = +125°C and ambient temperature of +55°C, the following steps guide the user in what the PCB pad size will need to be with 1 oz, 2 oz, and 3 oz copper.

a. Find size of copper pads on standard FR4 PCB to support operation at 1 amp $I_O$ square wave switching at a 0.50 duty factor (50 percent duty cycle) at $T_J$ = +125°C with $T_A$ = +55°C.

b. Calculate peak $I_F$ = 1A / 0.50 duty factor = 2 amps.

c. Use the $V_F$ versus $I_F$ curve on figure 8 to look up $I_F$ = 2 A (Y-axis) and follow across to the $T_J$ = +125°C curve (middle) for $V_F$ = 0.775 V.

d. Calculate power = $I_F$ x $V_F$ x duty factor = 2 x 0.775 x 0.50 = 0.775 W.

e. Calculate maximum thermal resistance needed (+125°C - 55°C) / 0.775 W = 90°C/W.

f. Look up thermal resistance of 90°C/W on the Y-axis using a thermal resistance versus pad area plot on one of the three curves on figure 7 for different weights of copper cladding and then intersect curve horizontally to get answer. Curves assume still air, horizontal position.

g. In this example, the copper mounting pad sizes for the different copper foil weights would be as follows:

(1) .092 $in^2$ (59.3 $mm^2$) for 1 ounce copper foil.
(2) .055 $in^2$ (35.5 $mm^2$) for 2 ounce copper foil.
(3) .036 $in^2$ (23.2 $mm^2$) for 3 ounce copper foil.

h. A conservative pad guard-band is optional since $T_J$ is only +125°C. NOTE: Multilayer PCBs, forced air cooling will improve performance. Closed confinement of the PCB will do the opposite. Use sound thermal management.



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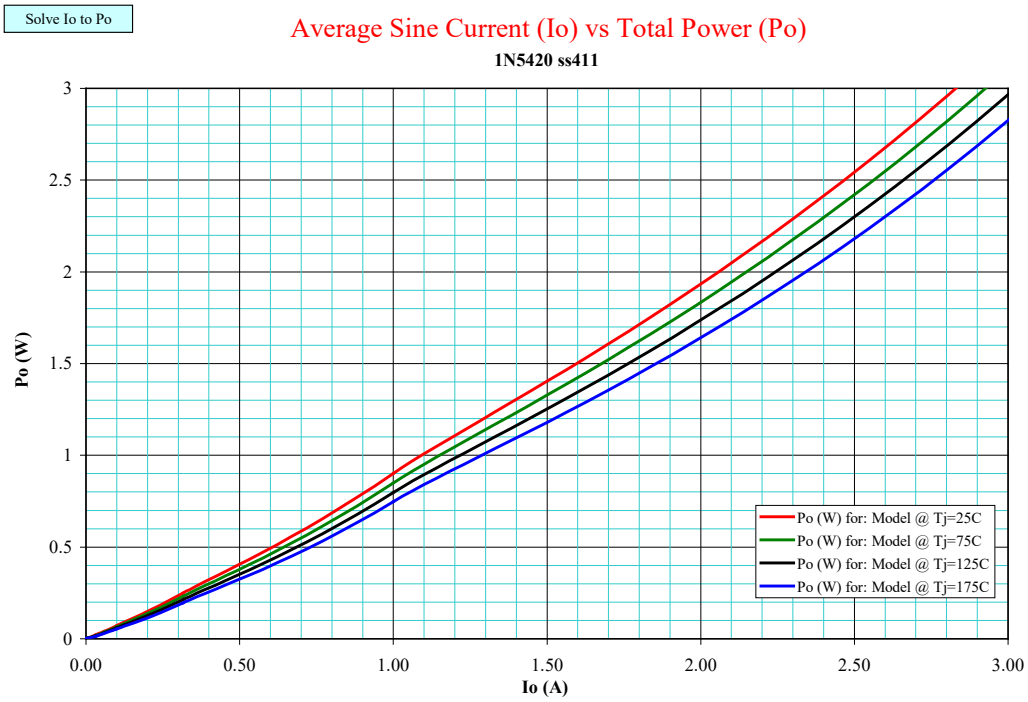



FIGURE 6. Rectifier power $P_O$ versus $I_o$ (average forward current).

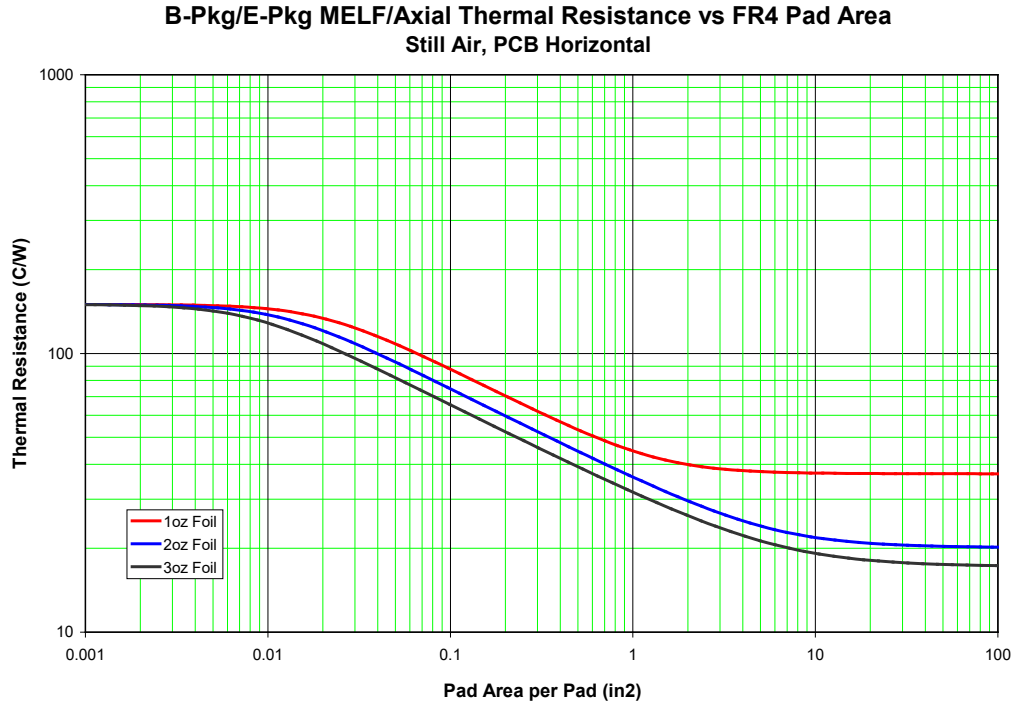



FIGURE 7. Thermal resistance versus pad area (for each pad) with 1 oz, 2 oz, and 3 oz copper.



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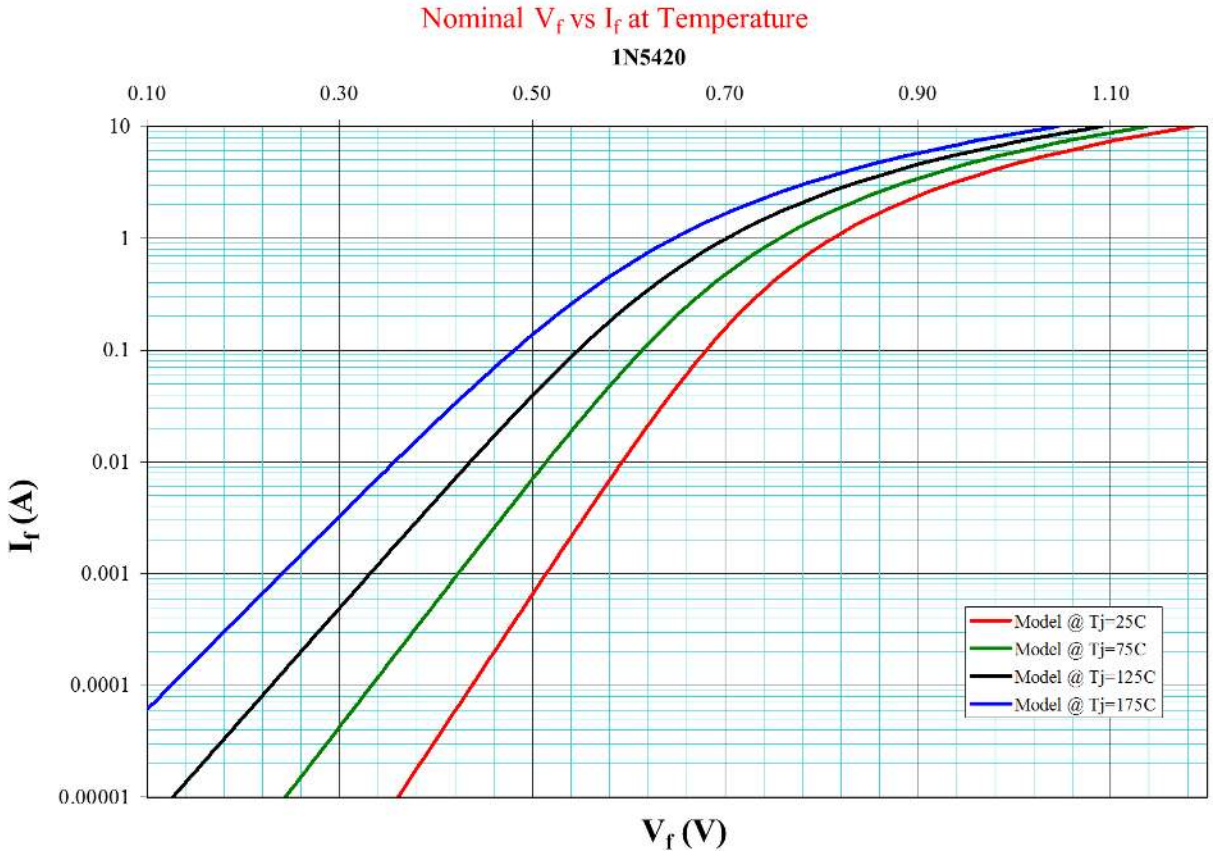



FIGURE 8. Forward voltage versus forward current for 1N5420.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.



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Preparing activity:
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